

Supporting Information

Benchmarking generative AI and physics based molecular simulation for sampling conformational heterogeneity of T4 Lysozyme

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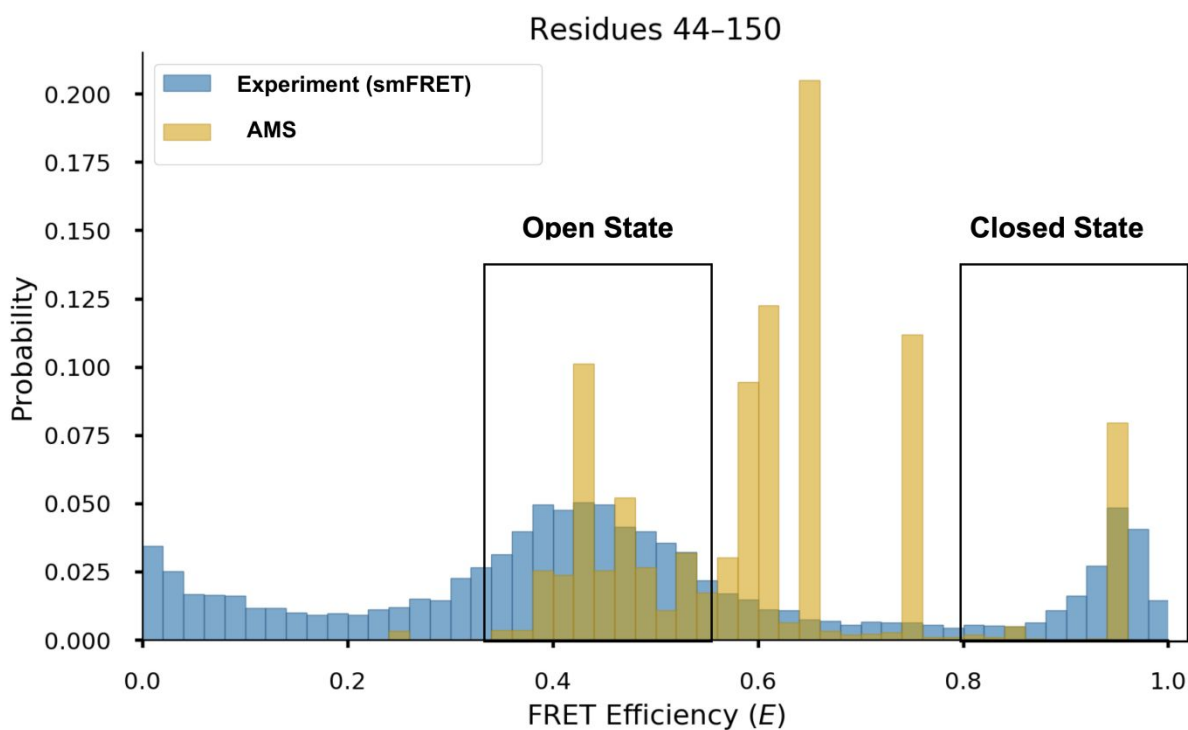


Figure S1. Comparison between MSM-weighted FRET efficiency from AMS and experimental smFRET shows sampling of both open and closed states in T4L. Notably, AMS captured the smFRET probability corresponding to the closed state, a rare event previously sampled only by EMD molecular simulations reported by *Bowman et al.*

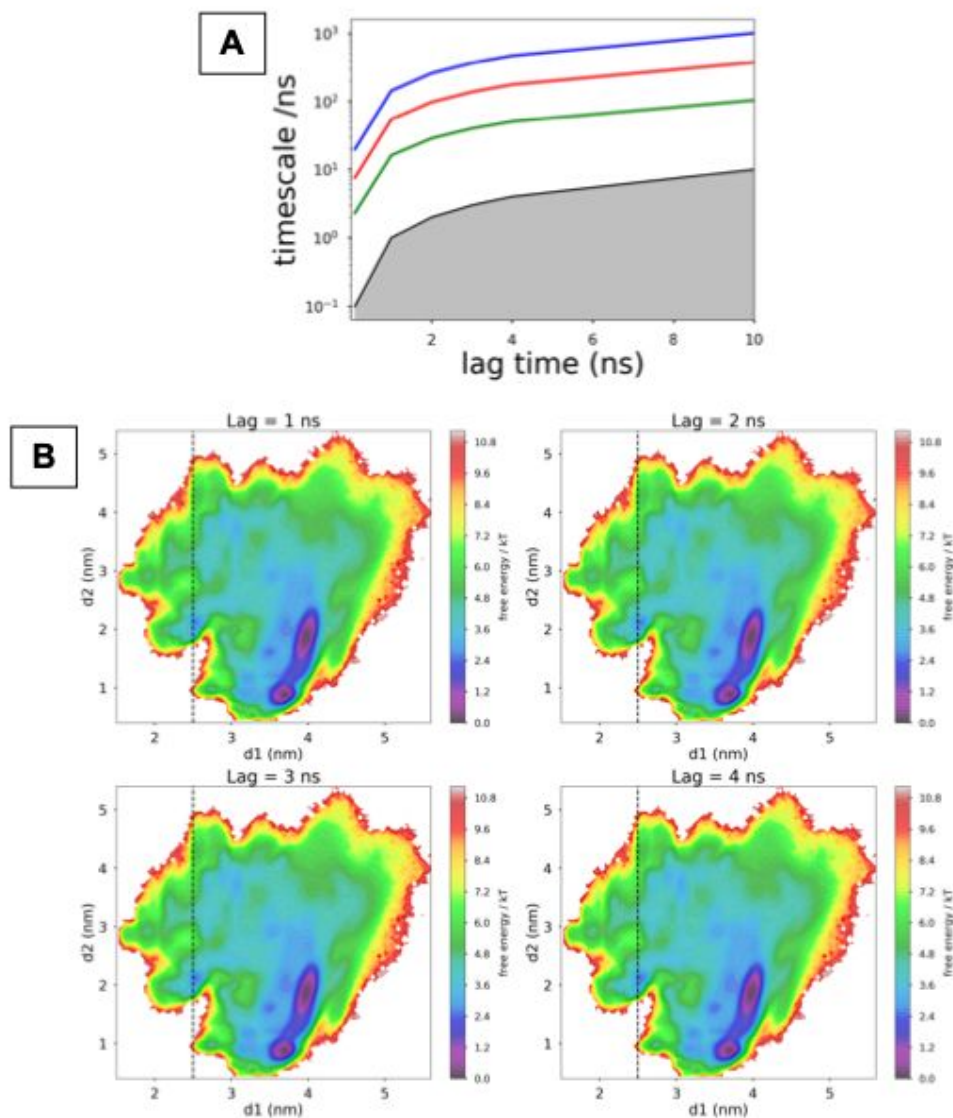


Figure S2. Lag time vs timescale plot and projection of MSM weighted free energy surface across different lag times highlighting convergence of AMS.

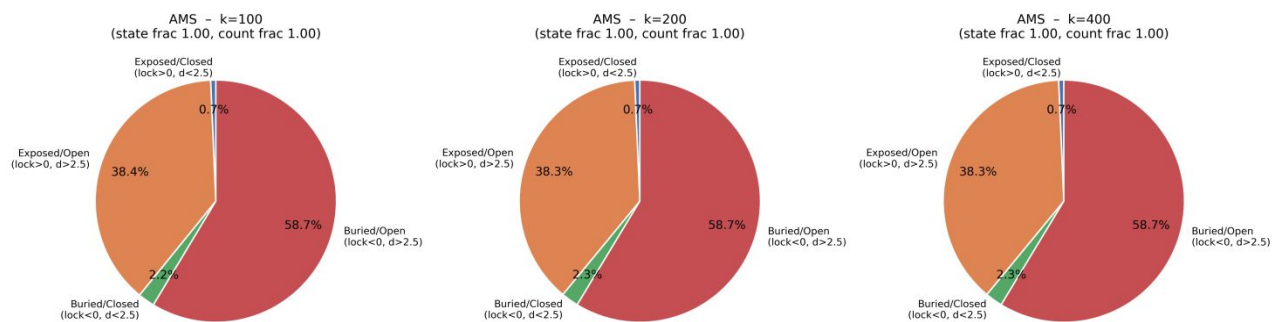


Figure S3. MSM-weighted equilibrium populations of distinct conformational states in wild-type apo T4L as a function of the number of cluster centers, K . The similar population distributions observed across different values of K indicate convergence of the AMS simulations.

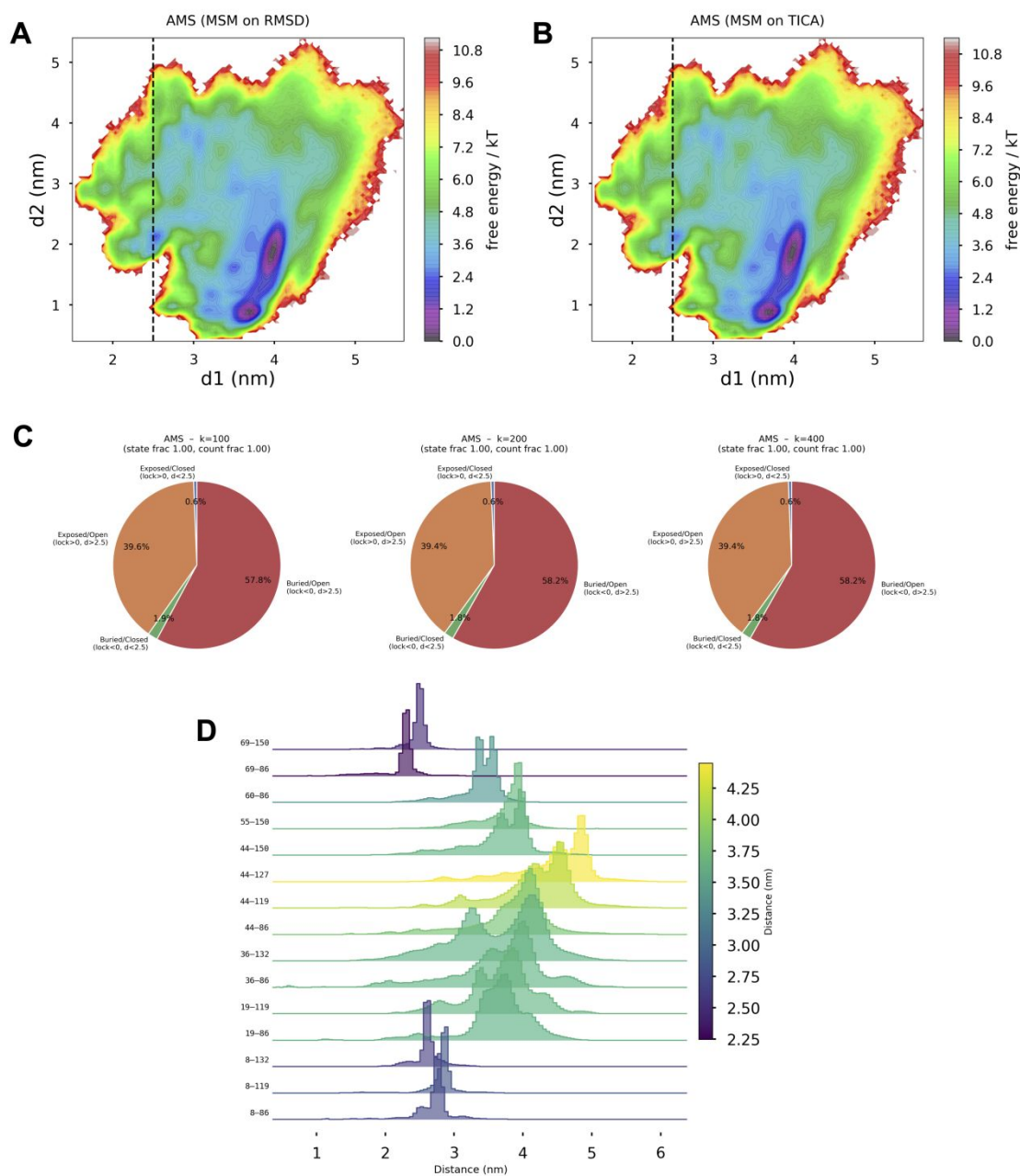


Figure S4. MSM-weighted free energy surfaces (FES) projected along $d1$ and $d2$ show consistent overall shape whether the underlying MSM is built from RMSD-based features (A) or from TICA components (lag time = 10) derived from pairwise C α -C α distance variables (B); the specific residue pairs used are listed in (D). These distance pairs were chosen because they correspond to the residue pairs for which authors reported a two-state experimental FRET distance distributions. K-means clustering of the TICA features across a range of cluster numbers ($k = 100, 200, 400$), followed by MSM estimation, further shows that the relative populations of the conformational states remain consistent across these choices of k (C).

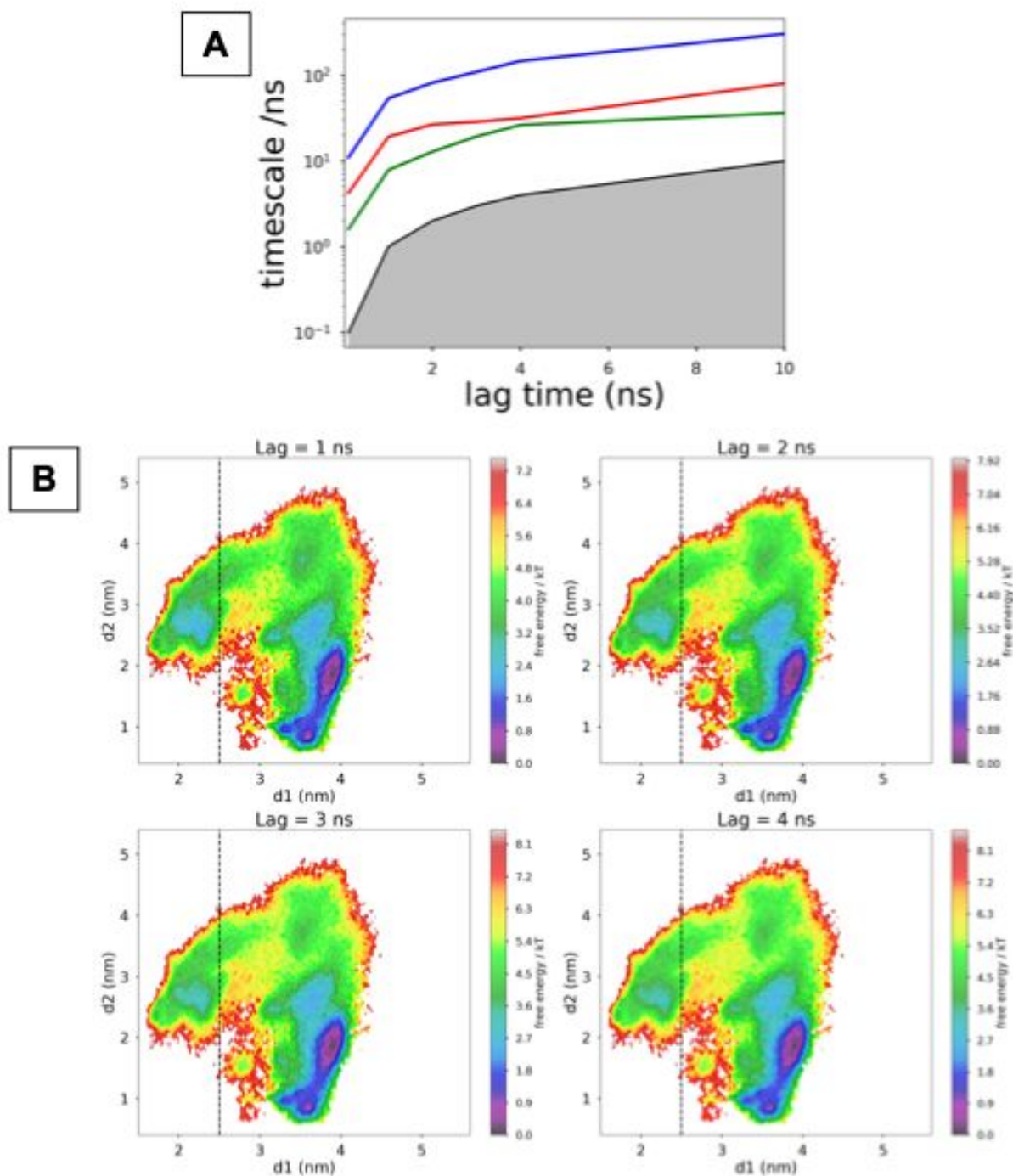


Figure S5. Lag time vs timescale plot and projection of MSM weighted free energy surface across different lag times highlighting the relative shape of the free energy surface in EMD simulations.

State	Population	95% CI
Exposed / Closed	0.7%	[0.6%, 0.8%]
Exposed / Open	38.3%	[37.3%, 38.9%]
Buried / Closed	2.3%	[1.7%, 3.3%]
Buried / Open	58.7%	[58.4%, 58.9%]

Figure S6. Bayesian MSM-weighted equilibrium populations (lag=500 steps, 5 ns in context of AMS simulation) and 95% confidence intervals for the four conformational states of wild-type apo T4L in AMS (Round 1 + Round 2). Link to Jupyter Notebook showing the statistics across different lag times can be accessed here <https://github.com/sbhakat/Gen-AI-t4l-bechmarking/blob/main/Analysis/stat-bayesian-msm.ipynb> :

State	Population	95% CI
Exposed / Closed	3.9%	[3.5%, 4.2%]
Exposed / Open	39.1%	[38.4%, 39.6%]
Buried / Closed	2.9%	[2.6%, 3.1%]
Buried / Open	54.0%	[53.4%, 55.0%]

Figure S7. Bayesian MSM-weighted equilibrium populations and 95% confidence intervals for the four conformational states of wild-type apo T4L in EMD.

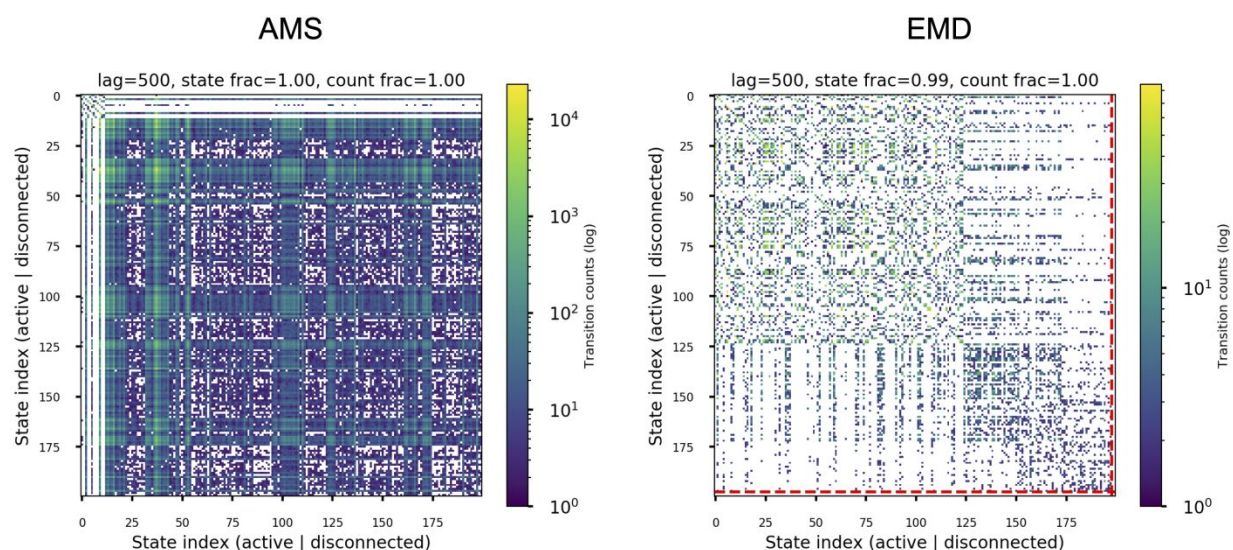


Figure S8. Transition count matrices (log-scale) for the AMS round 1 and EMD models at lag time = 500 steps and $K = 200$ (K-means clustering), with states reordered so that the largest connected component appears first, followed by any disconnected states (separated by a red dashed line where applicable). The EMD count matrix shows a distinct block of disconnected states with zero observed transitions to the main network (state fraction = 0.99, count fraction = 1.00), whereas the AMS count matrix is fully populated with no disconnected states (state fraction = 1.00, count fraction = 1.00), providing direct visual confirmation that AMS round 1 alone yields a fully connected MSM while EMD does not.

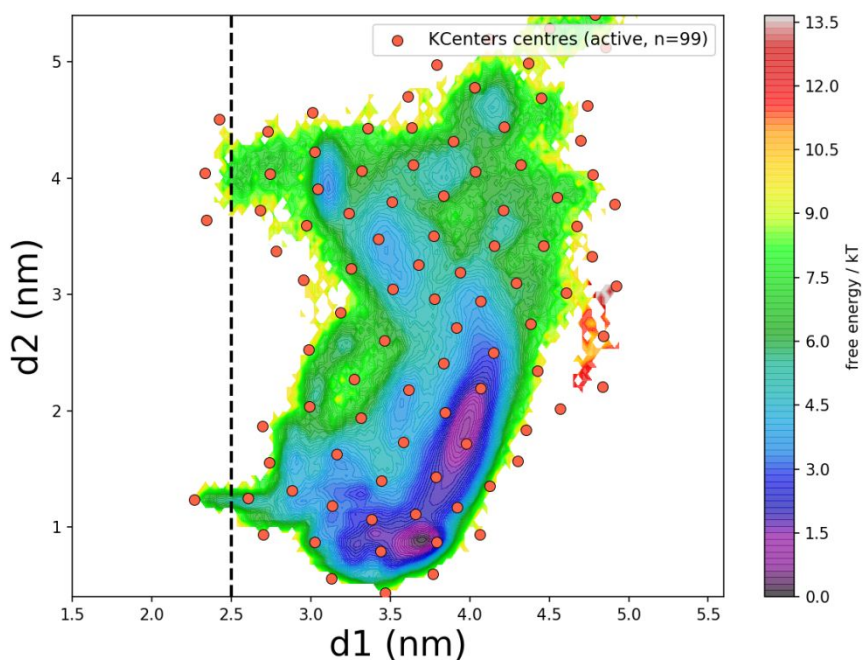


Figure S9. K-center clusters from AMS without AF-cluster projected on the 2D FES projected along d_1 and d_2 shows the ability of the algorithm in capturing conformational diversity which can be used as seeds to launch round 2 of AMS which had a high probability of sampling the closed state. The clustering was performed on the 2D conformational space characterized by d_1 and d_2 .